

# Fundamental Momentum Based Stock Investing

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# Background

## Extensive Research on Momentum Strategies

- Explaining the profitability of momentum strategies ([Chan et al., 1996](#))
- Exploring the effectiveness of momentum strategies across diverse asset classes ([Asness et al., 2013](#))
- Constructing momentum factor to describe the cross-sectional variation of commodity returns ([Bakshi et al., 2019](#))

# Motivation

## Recent Development in Momentum Strategies

- The traditional momentum is based on prices (i.e. **price momentum**).
- **Basis-momentum** factor proposed by [Boons and Prado \(2019\)](#) shows effective performance on predicting commodity premiums in both the time series and cross-section.
- **Macro momentum** strategies developed by [Brooks \(2017\)](#) demonstrates consistent performance over a half-century.
- **Fundamental momentum** strategy proposed by [Novy-Marx \(2015\)](#) provides more stable and significant performance compared with price momentum.

# Main Results

## Goal of Our Project

We aim to produce new fundamental momentum strategies based on the framework of [Novy-Marx \(2015\)](#), test the effectiveness of the strategy and how they relate to the traditional price momentum strategies, so that we can provide a more comprehensive understanding of momentum strategies.

## Empirical Interests

- Strategy performance in back-testing framework.
- Spanning tests.
- Relation with traditional price momentum strategies.
- Conditional strategies.

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# Data

## Data Sources: WRDS

- **Monthly Stock Data:** CRSP database.
- **Quarterly Company Data:** Compustat database.

## Data Process

- **Time Range:** from January 1975 to December 2023.
- Use CUSIP to match the stock price with the fundamental data (24260 stocks in the panel).
- Negative prices are dropped.

# Methodology: Backtesting Framework

## Quarterly-rebalanced Sorted Portfolio

- At each March, June, September, and December, we compute the fundamental momentum of the selected characteristics.
- If the momentum data is missing when we rebalance the portfolio, we use the most recent data up to two months.
- Considering the liquidity, we exclude the stocks with the lowest 30% volume.
- Sorting the stocks into deciles based on the fundamental momentum, we construct 10 sorted (equally-weighted) portfolios and hold for 3 months.

# Methodology: Profit Margin Momentum

## Why Profit Margin?

- Profit margin (PM) is a financial metric that indicates how much of a company's revenue remains as profit after covering all expenses.

$$PM = \frac{\text{Net Income}}{\text{Revenue}} \quad (1)$$

- A higher profit margin indicates a more profitable company that has better control over its costs compared to its competitors.

# Methodology: Profit Margin Momentum

## Factor Construction

$$\text{PM Momentum}(t) = \frac{PM(t) - PM(t - 3)}{SD(PM_{\text{Last } 3\text{Yrs}})} \quad (2)$$

- To increase the stability, we standardize the profit margin momentum by the standard deviation of the profit margin over the last 3 years.
- Moreover, we require that there are at least 6 observations in the last 3 years to calculate the standard deviation.

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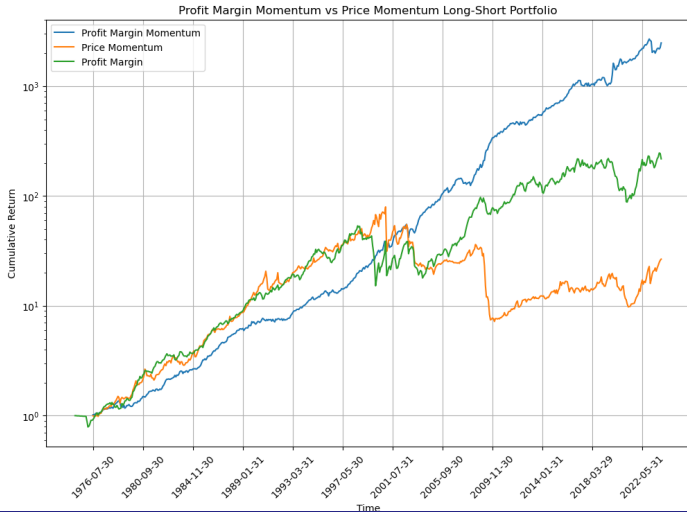
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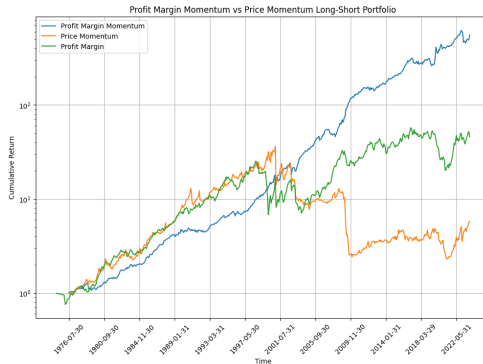
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# Sorted Portfolio based on Profit Margin Momentum



# Long-short Portfolio based on Profit Margin Momentum



|      | PM Momentum | Price Momentum | PM     |
|------|-------------|----------------|--------|
| Mean | 17.34%      | 10.17%         | 13.74% |
| Std  | 12.55%      | 23.67%         | 22.25% |
| SR   | 1.38        | 0.43           | 0.62   |
| MDD  | 25.70%      | 90.95%         | 71.53% |

# Spanning Test

## Regression Settings

- Regress profit margin momentum on FF3 factors and price momentum.
  - ▶ Examine whether the profit margin momentum can be subsumed.

## Regression Results: $R^2=2.60\%$

|             | $\alpha$ | MKT     | SMB    | HML    | MOM    |
|-------------|----------|---------|--------|--------|--------|
| Coefficient | 0.015    | -0.000  | 0.000  | 0.002  | 0.021  |
| t-statistic | (9.93)   | (-0.73) | (0.88) | (3.59) | (0.96) |

# Strategy Overlap: Definition

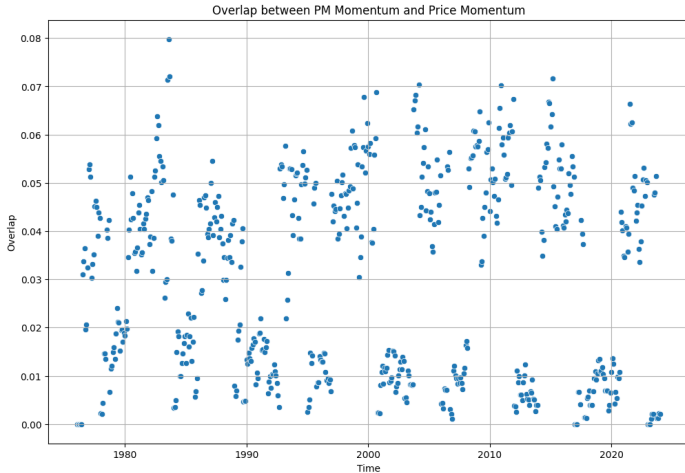
## Overlap Ratio

- At time  $t$ , for the  $i$ -th portfolio of profit margin momentum and price momentum, we calculate the overlap ratio simply by

$$\text{Overlap Ratio} = \frac{\# \text{ of Overlap Stocks}}{\# \text{ of Stocks (Union)}}$$

- For the long-short portfolios, we just take the average of the overlap ratio over the highest and lowest decile portfolios.

# Strategy Overlap: Results



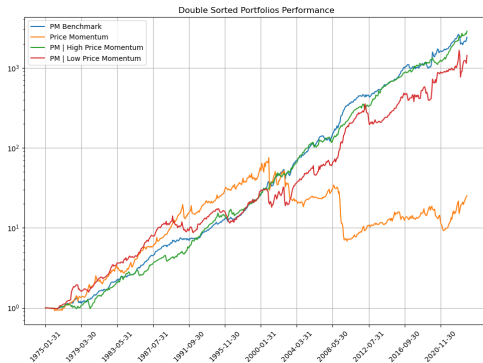
# Conditional Strategy

## What can be improved?

We want to control the effect of price momentum while building long-short profit margin momentum portfolio. We build long-short double sorted portfolios conditioning on PM momentum and price momentum. We compared the following two long-short portfolios.

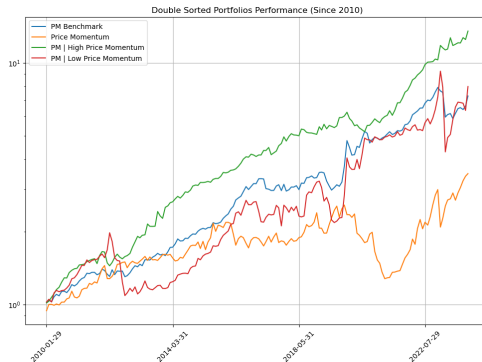
- High PM | High Price – Low PM | High Price
- High PM | Low Price – Low PM | Low Price

# Conditional Strategy: Results



|      | PM Momentum | High Price M. | Low Price M. |
|------|-------------|---------------|--------------|
| Mean | 16.76%      | 17.10%        | 17.40%       |
| Std  | 12.39%      | 12.15%        | 21.96%       |
| SR   | 1.35        | 1.41          | 0.79         |
| MDD  | 25.70%      | 23.39%        | 53.68%       |

# Conditional Strategy: Results(Since 2010)



|      | PM Momentum | High Price M. | Low Price M. |
|------|-------------|---------------|--------------|
| Mean | 15.39%      | 19.51%        | 19.36%       |
| Std  | 15.03%      | 12.41%        | 29.38%       |
| SR   | 1.02        | 1.57          | 0.66         |
| MDD  | 25.70%      | 16.95%        | 53.68%       |

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# Conclusion

- Profit margin momentum strategy provides a more stable and significant performance compared with price momentum.
- The profit margin momentum strategy is not subsumed by the traditional Fama-French 3 factors and price momentum.
- Though the overlap of the profit margin momentum and price momentum strategies is relatively low, the conditional strategy generates a more stable performance by reducing the effect of price momentum.

# Thank You!